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Mathematical & Theoretical Physics Notes

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Abstract : In an attempt to improve both my physics and english, this collection of personal notes aimed at my future self was created, covering theoretical physics and mathematical physics.
It will be a set of notes divided in two parts, in the first part, I will take notes of subjects studied this past academic year on math methods and in the second I'll take notes of self-studied more fundamental and pertinent subjects.

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Ideas for this project

Improve the accuracy of the 25-26 courses' « retranscription ».
Include weights and the exponential map, and maybe some axiomatic logic (Schuller's course ?)
Study QFT out of Weinberg's ad include talk on $SU(2)$, $SU(3)$ and the Poincaré group.

25-26

I Differential equations

Here I display my favourite equations and resolution methods, more can be found at [1].

1. Bernoulli's differential equation

Let us consider a Bernoulli differential equation defined as

$$\frac{\mathrm{d}}{\mathrm{d}x}y(x) = a(x)y(x) + b(x)y^n(x), \text{ where } n \in \mathbb{R}$$

rewritten as :

$$y^{-n}(x)\frac{\mathrm{d}}{\mathrm{d}x}y(x) = a(x)y^{1-n}(x) + b(x)$$

The resolution of such an equation begins by the substitution of the function $y(x)$ by a quite trivial function

$$u(x) = y^{1-n}(x)$$

such a substitution leads to a simpler equation which we know how to solve

$$\frac{1}{1-n}\frac{\mathrm{d}}{\mathrm{d}x}u(x) = a(x)u(x) + b(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}u(x) = p(x)u(x) + q(x)$$

2. Characteristic's equation

Let us consider the following equation :

$$a(x,y,z)\frac{\partial(x,y,z)}{\partial x}f + b(x,y,z)\frac{\partial(x,y,z)}{\partial y}f + c(x,y,z)\frac{\partial(x,y,z)}{\partial z}f = 0$$

We can define the equation above as the derivative of $f(x, y, z)$ by a variable along which it is constant

$$\frac{\mathrm{d}f}{\mathrm{d}s} = a(x,y,z)\frac{\partial(x,y,z)}{\partial x}f + b(x,y,z)\frac{\partial(x,y,z)}{\partial y}f + c(x,y,z)\frac{\partial(x,y,z)}{\partial z}f = 0$$

Leading to the so-simple equations

$$a(x,y,z) = \frac{\mathrm{d}x}{\mathrm{d}s} \text{ and } b(x,y) = \frac{\mathrm{d}y}{\mathrm{d}s} \text{ and } c(x,y) = \frac{\mathrm{d}z}{\mathrm{d}s}$$

Naturally the next to be taken to determine $s(x, y, z)$ is to isolate $\mathrm{d}s(x, y, z)$

$$\mathrm{d}s(x, y, z) = \frac{\mathrm{d}x}{a(x, y, z)} = \frac{\mathrm{d}y}{b(x, y, z)} = \frac{\mathrm{d}z}{c(x, y, z)}$$

By isolating the three RHS [1] terms 2 by 2 (one of the combinations may be omitted as it carries redundant information), we find

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{b(x, y, z)}{a(x, y, z)} \text{ and } \frac{\mathrm{d}y}{\mathrm{d}z} = \frac{b(x, y, z)}{c(x, y, z)}$$

and the solutions to these differential equations are none other than two families of characteristic lines along wich $f(x, y)$ is conserved. These sets of solutions are sets that depend on the integration constants that shall be determined using boundary conditions [2].

Finally [3] :

$$f(x, y, z) = g(\Psi(x, y, z), \Phi(x, y, z))$$

3. Wronskian

When studying a physical problem, it is interestig to find the family of functions that are solution to a category of Differential equation. The use of the Wronskian allows to determine (starting from $n - 1$ sets of solutions) the last set of solutions to an n-th order linear homogeneous differential equation, not any set of solutions but a linearly independent family that would complete the $n - 1$ collection of families leading to us having the "basis" of solutions to n order linear homogeneous differential equations. Let us consider this equation

$$f^{(n)}(x) + p_{n-1}(x)f^{(n-1)}(x) + \cdots + p_1(x)f'(x) + p_0(x)f(x) = 0$$

The Wronskian is the following determinant

$$W(f_1, f_2, \ldots, f_n)(x) = \begin{vmatrix} f_1(x) & f_2(x) & \cdots & f_n(x) \\ f_1'(x) & f_2'(x) & \cdots & f_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \cdots & f_n^{(n-1)}(x) \end{vmatrix}$$

According to Abel's [4] identity

$$\frac{\mathrm{d}W}{\mathrm{d}x} = -p_{n-1}(x)W(x) \rightarrow W(x) = C \exp\left(-\int p_{n-1}(x)\frac{\mathrm{d}}{\mathrm{d}x}\right)$$

1. Right Hand Side.
2. Dirichlet boundary conditions specify conditions on the solution itself, whereas Neumann boundary conditions specify conditions on the derivatives of the solution, the combinations of these conditions are diverse and may be found on `https://en.wikipedia.org/wiki/Mixed_boundary_condition`.
3. this method can be applied at to an N-variable function but I only explained the three variable one as the difference between each N-dimensional implementation of it is that we only generate $N - 1$ equations thus $N - 1$ characteristic sets of lines.
4. what a Man !

By develloping the determinant on the last column we obtain :

$$\sum_{k=0}^{n-1} M_k(x) f_n^{(k)}(x) = C \exp\left(-\int p_{n-1}(x) \frac{\mathrm{d}}{\mathrm{d}x}\right)$$

Which should in theory be solvable.

4. Laplace Problem

Laplace's problem[5] is famous, elegant, useful and omnipresent in the physics I've encountered up until now. Be it in Electromagnetism, elementary Gravity problems, and of course in the use of Green's functions.

$$\Delta f = 0$$

the 1D problem is trivial, so are the 2D and 3D ones to be fair, yet … here we go :

α. 1-dimensional problem

$$\frac{\partial^2 f}{\partial x^2}(x) = 0 \implies f(x) = Ax + B, \text{ where } A, B \in \mathbb{R}$$

β. 2-dimensional problem

$$(\frac{\partial^2 +}{\partial x^2}\frac{\partial^2)}{\partial y^2} f(x, y) = 0 \iff \frac{\partial^2 (x, y)}{\partial x^2} f = -\frac{\partial^2 (x, y)}{\partial y^2} f$$

and as we can, we will decompose $f(x, y) = X(x)Y(y)$

$$\frac{\frac{\partial^2 X}{\partial x^2}(x)}{X(x)} = -\frac{\frac{\partial^2 Y}{\partial y^2}(y)}{Y(y)}$$

and as both the LHS and the RHS are linearly independent because of their different variables, we can therefore guess that they are equal to a constant K, whose real or imaginary nature will influence the results we get.

γ. 3-dimensional problem

The 3-dimensional version of the above problems are almost exactly the same, only the number of of solutions obtained varies. The idea of the laplace problem resolution is centered around variable separation and if needed more than one arbitrary constant may be used to dully seperate the variable.

5. For further information and/or details check out the *GOAT*'s lecture notes : Wietze HERREMAN.

5. Green's functions

Green's method is a way to solve a problem through the definition of what is called a Green function $G(\vec{r}-\tilde{\vec{r}})$ which holds a significant physical relevance. The first step consists of the rewritting of our n-dimensional physical equation as integrals

$$\Delta\phi(\vec{r}-\tilde{\vec{r}}) = k\rho(\vec{r}-\tilde{\vec{r}}) \longrightarrow \int G(\vec{r}-\tilde{\vec{r}})\rho(\tilde{\vec{r}})\mathrm{d}\tilde{r} = k\int \delta^n(\vec{r}-\tilde{\vec{r}})\rho(\tilde{\vec{r}})\mathrm{d}\tilde{r}$$

This leads to the much simpler equation

$$\Delta G(\vec{r}-\tilde{\vec{r}}) = k\delta^n(\vec{r}-\tilde{\vec{r}})$$

It is interesting to notice that if for example we take ϕ a gravitational potential for a punctual star of mass M located at the origin of our frame, we therefore have[6] $k = 4\pi GM$[7] and $\rho(\vec{r}) = \delta^3(\vec{r})$.

α. 1-dimensional problem

A one dimensionnal Green function problem gets simplified to an equation of the form :

$$\frac{\partial^2 G}{\partial r^2}(r-\tilde{r}) = \delta(r-\tilde{r})$$

We first solve the previous equation for $x \neq 0$

$$G_\pm\big|_{r\neq\tilde{r}} = C_\pm\,(r-\tilde{r}) + B_\pm, \text{ with : } C_\pm, B_\pm \in \mathbb{R}$$

Boundary conditions allow us to determine either (or both) integration constant(s), and another way of determining the hypothetical undertermined integration constant is through the integration over the parameter of the dirac delta.

$$\int_\gamma \frac{\partial^2 G}{\partial r^2}(r-\tilde{r})\mathrm{d}r = \int_\gamma \delta(r-\tilde{r})\mathrm{d}r, \text{ where } \gamma\ :\ r\in[-\varepsilon;\varepsilon]$$

$$\frac{\partial G}{\partial r}(\varepsilon) - \frac{\partial G}{\partial r}(-\varepsilon) = C_+ - C_- = 1$$

and using further conditions and information on our physical system it is possible to determine the constants $C_\pm$ and $B_\pm$. This cas is very specific as the absence of a solution due to an "infinite density" at $r = \tilde{r}$ makes the r axis disconnected, and so there is no reason to make the terms at $\tilde{r}^-$ and $\tilde{r}^+$ be the same.

β. 2-dimensional problem

A two dimensionnal Green function problem[8] gets simplified to an equation of the form :

$$\Delta G(\vec{r}) = r^{-1}\frac{\partial}{\partial r}\left(r\frac{\partial G}{\partial r}\right) = k\delta^2(\vec{r})$$

6. the G used in k stands for the universal gravitational constant.
7. in the case of other physical theories such as Electricity or Magnetism, the constants k differ (q/ε_0 in case of the scalar electric potential).
8. in cylindrical coordinates.

Solving this problem as a Laplace problem first

$$G = C \ln\left(|\vec{r} - \tilde{\vec{r}}|\right) + B$$

then integrating over a constant r surface (a circle)

$$\int \Delta G \mathrm{d}^2 S = \int k\delta^2(\vec{r} - \tilde{\vec{r}})\mathrm{d}^2 S = k$$

$$k = \oint R \mathrm{d}\theta \left[\vec{\nabla}\, G|_{r=R} \cdot \vec{e}_r\right] = 2\pi C$$

$$C = \frac{k}{2\pi}$$

Finally using the boundary conditions as $r \to \infty$ we determine the value of B

$$G(\vec{r} - \vec{\tilde{r}}) = B + k\frac{\ln|\vec{r} - \tilde{\vec{r}}|}{2\pi}$$

γ. 3-dimensional problem

A three dimensional Green problem simplifies to

$$\Delta G(\vec{r} - \tilde{\vec{r}}) = k\delta^3(\vec{r} - \tilde{\vec{r}})$$

once again, let us solve the homogeneous equation where $\vec{r} \neq \tilde{\vec{r}}$:

$$r^{-2}\frac{\partial}{\partial r}\left(r^2\frac{\partial G}{\partial r}\right) = 0 \implies G(\vec{r} - \tilde{\vec{r}}) = -\frac{C}{\|\vec{r} - \tilde{\vec{r}}\|} + B$$

We input this in the volume integral

$$\int_V \Delta G \mathrm{d}V = \oint_{\partial V} \vec{\nabla} G \cdot \mathrm{d}\vec{S}$$

$$C = \frac{k}{4\pi}$$

whereas B is determined by our G value at $\|\vec{r} - \tilde{\vec{r}}\| \to \infty$

$$G(\vec{r} - \tilde{\vec{r}}) = -\frac{k}{4\pi\|\vec{r} - \tilde{\vec{r}}\|} + B$$

Applying these results to our previous example regarding the star of mass M (in section 5.), we get

$$\phi(\vec{r}) = -\int \left[\frac{GM}{\|\vec{r} - \tilde{\vec{r}}\|} - B\right]\delta^3(\tilde{\vec{r}})\mathrm{d}\tilde{r}$$

leading to the well-known result[9] :

$$\phi(\vec{r}) = -\frac{GM}{\|\vec{r}\|} + B$$

9. still for a punctual mass M positionned at the center of the frame.

II Complex analysis

Complex analysis [1] is the study of functions of complex variables. To put it simply we will focus on functions such as $f(z) \in \mathbb{C}$ with $z \in \mathbb{C}$.

1. Generalities

Some general notations commonly used

$$f(z) = f(x, y) = u(x, y) + \mathbf{i}\, v(x, y), \text{ where} : u, v \in \mathbb{R}$$

$f(z)$ is a **holomorphic** function in Ω an open subset of $\mathbb{C}$ if the equation above holds $\forall z \in \Omega$. It is important when doing analysis to have good definitions of derivation. It just so happens that for a matter of consistency we define it the same way we did in real analysis (we simply replace x by z), only we add the condition that the usual (following) equality must hold no matter the path taken

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h} = \frac{\mathrm{d}f(z)}{\mathrm{d}z}$$

Therefore if we take a path along either x or iy, we must have (for functions we will call **analytical**) equality of the derivatives along both paths :

$$f'(z_0) = \lim_{x \to x_0} \frac{u(x, y_0) - u(x_0, y_0)}{x - x_0} + \mathbf{i} \lim_{x \to x_0} \frac{v(x, y_0) - v(x_0, y_0)}{x - x_0} = \frac{\partial u(x, y)}{\partial x} + \mathbf{i}\, \frac{\partial v(x, y)}{\partial x}$$
$$f'(z_0) = \lim_{y \to y_0} \frac{u(x_0, y) - u(x_0, y_0)}{\mathbf{i}\, y - \mathbf{i}\, y_0} + \mathbf{i} \lim_{y \to y_0} \frac{v(x_0, y) - v(x_0, y_0)}{\mathbf{i}\, y - \mathbf{i}\, y_0} = -\mathbf{i}\, \frac{\partial u(x, y)}{\partial y} + \frac{\partial v(x, y)}{\partial y}$$

Leading to the well-known Cauchy-Riemann (CR) conditions of analyticity for a complex function

$$\boxed{\frac{\partial u(x, y)}{\partial x} = \frac{\partial v(x, y)}{\partial y} \qquad \& \qquad \frac{\partial u(x, y)}{\partial y} = -\frac{\partial v(x, y)}{\partial x}} \tag{1.1}$$

We can define the derivatives

$$\partial_z = \frac{1}{2}\left(\frac{\partial}{\partial x} - \mathbf{i}\, \frac{\partial}{\partial y}\right) \qquad \text{and} \qquad \partial_{\bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + \mathbf{i}\, \frac{\partial}{\partial y}\right)$$

1. [2] is a good ressource to study Complex Analysis.

leading to the re-writing of the CR conditions as

$$\frac{\partial f}{\partial \bar{z}} = 0$$

We call **analytical** function any function that can be decomposed into a Laurent series

$$f(z) = \sum_{n=-\infty}^{\infty} a_n \left(z - z_0\right)^n = \sum_{n=-\infty}^{-1} a_n \left(z - z_0\right)^n + \sum_{n=0}^{\infty} a_n \left(z - z_0\right)^n$$

we will refer to the LH term the singular part of $f(z)$ and the RH term the analytical part of $f(z)$.

A **meromorphic** function is one such that the derivative is defined on $\left(\Omega \setminus \left\{z_i\right\}_{i\in\mathbb{Z}}\right) \subset \mathbb{C}$, implying it isn't derivable on $\left\{z_i\right\}_{i\in\mathbb{Z}}$ and we call this set of points **singular points** (basically singularities). There are three kinds of singularities :

- ***n*-th order pole**, which is characterized by $f(z) \propto \left(z - z_0\right)^{-n}$
- **Essential** singularities, characterized by infinite terms proportional to $\left(z - z_0\right)^{-1}$, and therefore a non vanishing singular part of the Laurent series decomposition of the function.
- **Apparent** singularities that aren't actually singularities.

2. Integration

A few results that are very practical to physicists. I omitted a few to speed my writing up and get to more interesting and relevant subjects faster but I will come back to this section.

Cauchy's Theorem :

When integrating a holomorphic function over a closed contour, it is expected because of the Cauchy theorem for its value to be null.

$$\oint_\Gamma \mathrm{d}z f(z) = 0$$

The holomorphy condition means the absence of singularities inside the contour as no singularities of $f(z)$ exist on Ω anyways. The contour containing no singularities is therefore called homotopic to 0.

Cauchy's generalized Theorem :

When integrating a meromorphic function over a closed contour homotopic to zero, it is expected because of Cauchy's generalized theorem for its value to be :

$$\oint_\Gamma \mathrm{d}z f(z) = \sum_i \oint_{\Gamma_i} \mathrm{d}z f(z)$$

where each Γ_i is a contour around one of the singularities contained in Γ.

Jordan's lemma :

Let us consider a circular contour $C_\mathcal{R}$ (not necessarily closed) then the following equation holds

$$\lim_{R\to 0} \mathrm{Sup}_{z\in C_\mathcal{R}} |zf(z)| = 0 \implies \lim_{R\to 0} \int_{C_\mathcal{R}} \mathrm{d}z f(z) = 0$$
$$\lim_{R\to \infty} \mathrm{Sup}_{z\in C_\mathcal{R}} |zf(z)| = 0 \implies \lim_{R\to \infty} \int_{C_\mathcal{R}} \mathrm{d}z f(z) = 0$$

This lemma is very practical for the determination of integrals using the upcoming *Residue Theorem.*

Residue Theorem :

Let $f(z)$ be a meromorphic function on $\Omega \subset \mathbb{C}$, let Γ be a closed contour surrounding N singularities of $f(z)$, and Γ doesn't cut through any singuarities of $f(z)$. The integral

$$\oint_{\Gamma>0} \mathrm{d}z f(z) = 2\mathbf{i}\,\pi \sum_{i=1}^{N} \mathrm{Res}\left(f(z), z_i\right)$$

where $\mathrm{Res}\left(f(z), z_i\right) = a_{-1}$, the term proportional to $\left(z - z_i\right)^{-1}$. For n-th order poles, there is another residue formula :

$$\mathrm{Res}\left(f(z), z_i\right) = \lim_{z\to z_i} \frac{1}{(n-1)!} \frac{\mathrm{d}^{n-1}}{\mathrm{d}z^{n-1}} \left[(z-z_i)^n f(z)\right]$$

Lebesgue's dominated convergence theorem :

Let $f_n(z)$ be a sequence of measurable functions, assuming it converges pointwise (simplement) almost everywhere to a function $f(z)$ on a measurable set A. If there exists a function $g(z)$ that is inetgrable (sommable) over A such that $|f_n(z)| \leq g(z) \quad \forall n \in \mathbb{N}$ almost everywhere, then :

$$\lim_{n\to\infty} \int_A \mathrm{d}z f_n(z) = \int_A \mathrm{d}z f(z), \text{where } f(z) = \lim_{n\to\infty} f_n(z)$$

Self-study

III Group Theory and representations

This section is heavily inspired by ([3],[4], [5] and [6])[1] which served as my primary self-study material[2].

1. On Groups

Group :
A group $(\mathcal{G}, \circ)$ is a set $\mathcal{G}$ equipped with an internal composition law "$\circ$", that respects axioms of :

$i)$ Associativity : $(a \circ b) \circ c = a \circ (b \circ c)$

$ii)$ Existence of a neutral element : $a \circ e = e \circ a = a$

$iii)$ Existence of an inverse element : $a \circ a^{-1} = e$

where obviously $a, b, c, a^{-1}, e \in \mathcal{G}$.
Note that commutativity isn't a requirement and that if a group has a commutative operation it is said to be **abelian**.

Topological Group :
A topological group is a topological space that is also a group, such that the group operation $\circ$ is continuous, same goes for the inversion mapping

Homomorphism :
A homomorphism φ is a mapping from a group $(\mathcal{G}, \circ)$ to a group $(\mathcal{H}, \star)$, such that

$$\varphi(a \circ b) = \varphi(a) \star \varphi(b), \quad \forall a, b \in \mathcal{G}$$

If said homomorphism is bijective, then it is also an **isomorphism**.
If $\mathcal{H} = \mathcal{G}$ and φ is bijective, it is then said to be an **automorphism**.
An **inner automorphism** is a mapping of the form $g' \rightarrow g g' g^{-1}$.

Center :
The center of a group $(\mathcal{G}, \circ)$ is the set of elements that commute with all other elements of $(\mathcal{G}, \circ)$:

$$Z_{\mathcal{G}} = \{g \in \mathcal{G} | g \circ a = a \circ g, \ \forall a \in \mathcal{G}\}$$

1. Especially [3]
2. [5, 6] are especially good respectively for problems to solve and physics exmaples.

From this point forward I will often be referring to $(\mathcal{G}, \circ)$ as "the group $\mathcal{G}$".

Subgroup :

A subgroup $\mathcal{H}$ of $\mathcal{G}$, is a subset of $\mathcal{G}$ that is stable under the internal composition law defined on $\mathcal{G}$. Simply put, it is a subset of $\mathcal{G}$ that is also a group under the same law.

Cosets :

Let us consider $\mathcal{H}$ a subgroup of $\mathcal{G}$. let us consider the effect "on the left" of $a \in \mathcal{G}$ on $\mathcal{H}$.

$$a\mathcal{H} = \{a \circ h \,|\, \forall h \in \mathcal{H}\}$$

$a\mathcal{H}$ is called a left coset of $\mathcal{G}$.
We can also define $\mathcal{G}/\mathcal{H}$ the group of cosets of $\mathcal{H} \subset \mathcal{G}$, also called quotient group of $\mathcal{G}$ by $\mathcal{H}$.

Invariant/normal subgroup :

A subgroup $\mathcal{H}$ of $\mathcal{G}$ is said to be invariant if $a\mathcal{H}a^{-1} = \mathcal{H}, \quad \forall a \in \mathcal{G}$.
By simply multiplying both sides of the previous equation with a to the right we show that right and left cosets are equal if $\mathcal{H}$ is invariant (normal).

$$a\mathcal{H}a^{-1}a = a\mathcal{H} = \mathcal{H}a$$

Consequently, if $\mathcal{H}$ is invariant, its left and right coset coincide.
In the context of the definition of $\mathcal{G}/\mathcal{H}$, it is interesting to note that the $\mathcal{H}$s in question are invariant subgroups, to see why let us consider two elements of $\mathcal{G}/\mathcal{H}$

$$(a\mathcal{H})\,(b\mathcal{H}) = (ab)\mathcal{H}, \quad \text{where } a, b \in \mathcal{G}$$

We must have in order for this equality to be true $\mathcal{H}b = b\mathcal{H}$ so that

$$(a\mathcal{H})\,(b\mathcal{H}) = a(\mathcal{H}b)\mathcal{H} = (ab)\mathcal{H}\mathcal{H} = (ab)\mathcal{H}$$

for clarity $\mathcal{H}\mathcal{H} = \mathcal{H}$ because of the closure of the group structure.

Also, card$\,(G)$ is the number of elements in $\mathcal{G}$, in case it is (in)finite $\mathcal{G}$ would then be called a(n) (in)finite group.
The **kernel**[3] of a homomorphism φ from $\mathcal{G}$ to $\mathcal{G}'$ is the set of elements of $\mathcal{G}$ that get mapped to the identity of $\mathcal{G}'$

$$\ker(\varphi) = \left\{g \in \mathcal{G} \,|\, \varphi(g) = e_{\mathcal{G}'}\right\}$$

It is straightforward to show, using arguments similar to those used for quotient groups, that the kernel of a homomorphism is an invariant subgroup of the domain.
Let $n \in \ker(\varphi)$, and $g, g^{-1} \in \mathcal{G}$

$$\varphi(gng^{-1}) = \varphi(g)\varphi(n)\varphi(g^{-1}) = \varphi(g)\varphi(g^{-1})e_{\mathcal{G}'} = e_{\mathcal{G}'}$$

3. Some of the definitions and notions I introduce seem disconnected from Physics, they are not. Take conjugacy classes for example, across one traces/characters are the same, and so are some other invariants that may be observables (from what I've picked up up until one at least).

Therefore $gng^{-1} \in \ker(\varphi)$, put differently :

$$\ker(\varphi) = g\ker(\varphi)g^{-1} \qquad \forall g \in \mathcal{G}.$$

The **conjugacy class** of $x \in \mathcal{G}$ is the set of $gxg^{-1} \in \mathcal{G},\ \forall g \in \mathcal{G}$.

2. On Representations

Representation :

A representation $(\mathcal{E}, U)$ of the group $\mathcal{G}$ on the vector space $\mathcal{E}$ is a homomorphism :

$$\begin{aligned} U\ :\mathcal{G} &\to \mathrm{GL}(\mathcal{E}) \\ g &\mapsto U(g) \end{aligned}$$

If said homomorphism is an isomorphism then the representation is said to be faithful.
The mapping $g \mapsto U(g)$ must be continuous if $\mathcal{G}$ is a topological group and $\mathcal{E}$ is a topological vector space.

We shall call the vector space $\mathcal{E}$ (as do some of the authors in the referenced textbooks) the **representation space** of $\mathcal{G}$, often called representation for simplicity. The dimension of the representation is that of the vector space it acts on, naturally. In the case of unitarity of the representation of each element of $\mathcal{G}$, i.e. $U^\dagger(g) = U^{-1}(g),\ \forall g \in \mathcal{G}$, it is said that the representation is **unitary**.
Choosing a specific basis of $\mathcal{E}$, the operators $U(g)$ are represented by matrix elements denoted $u^i_{\ j}(g)$ for which the following equation holds [4] :

$$u^i_{\ j}(g) = u^i_{\ k}(g)u^k_{\ j}(g)$$

Character :

The characters χ_U of a representation of $\mathcal{G}$ are the traces of the matrix representations

$$\mathrm{tr}\, U(g) \equiv u^i_{\ i}(g)$$

Let $(\mathcal{E}_1, U_1)$ and $(\mathcal{E}_2, U_2)$ of the same group $(\mathcal{G}, \circ)$ [5] are said to be equivalent if there exists an isomorphism T of $\mathcal{E}_1$ on $\mathcal{E}_2$ such that :

$$TU_1(g) = U_2(g)T \qquad\qquad \forall g \in \mathcal{G}$$

The **direct sum** of two representations $\left(\mathcal{E}_1, U_1\right)$ and $\left(\mathcal{E}_2, U_2\right)$ of the same group $(\mathcal{G}, \circ)$ has $\mathcal{E}_1 \oplus \mathcal{E}_2$ as the representation space, and $U_1 \oplus U_2$ as operators

$$\begin{aligned} U_1 \oplus U_2\ :\ \mathcal{E}_1 \oplus \mathcal{E}_2 &\to \mathcal{E}_1 \oplus \mathcal{E}_2 \\ \psi_1 \oplus \psi_2 &\mapsto U_1(g)\psi_1 \oplus U_2(g)\psi_2 \end{aligned}$$

4. Einstein's summation convention is used throughout these notes !
5. Yeah I just wrote it rigorously, surprised ?

In matrix notation, we can find a basis in which $U(g)$ is represented by

$$u(g) = \begin{pmatrix} u_1(g) & 0 \\ 0 & u_2(g) \end{pmatrix}$$

The character of a direct sum of representations is the sum of the characters of each representation of the direct sum.

The **direct product** of two groups $(\mathcal{G}_1, \circ)$ and $(\mathcal{G}_2, \star)$ is the group $(\mathcal{G}_1 \times \mathcal{G}_2, \cdot)$ with group operation such that $(a_1, b_1) \cdot (a_2, b_2) = (a_1 \circ a_2, b_1 \star b_2)$.

The **tensor product** of two representations $\left(\mathcal{E}_1, U_1\right)$ and $\left(\mathcal{E}_2, U_2\right)$ has $\mathcal{E}_1 \otimes \mathcal{E}_2$ as the representation space, and $U_1 \otimes U_2$ as operators

$$\begin{aligned} U_1 \otimes U_2 : \mathcal{E}_1 \otimes \mathcal{E}_2 &\to \mathcal{E}_1 \otimes \mathcal{E}_2 \\ \psi_1 \otimes \psi_2 &\mapsto U_1(g)\psi_1 \otimes U_2(g)\psi_2 \end{aligned}$$

In matrix notation, $u_{(ij)(km)}(g) = u_{1\ ik}(g) u_{2\ jm}(g)$. The characters of a tensor product are the products of the characters of the representations of the product.

Reducible representation :

A representation $(\mathcal{E}, U)$ is said to be reducible if there exists $\mathcal{E}_1 \subset \mathcal{E}$ such that

$$U(g)\mathcal{E}_1 \subseteq \mathcal{E}_1, \qquad \forall g \in \mathcal{G}$$

written differently

$$U(g)\psi \in \mathcal{E}_1; \qquad \forall \psi \in \mathcal{E}_1\ ,\ \forall g \in \mathcal{G}$$

when put in matrix notation, it means there exists a basis in which the action on a subspace of $\mathcal{E}$, namely $\mathcal{E}_1$ is

$$u(g) = \begin{pmatrix} u_1(g) & x(g) \\ 0 & y(g) \end{pmatrix}$$

An **irreducible representation** is a representation that isn't reducible[6].

Completly reducible representation :

A representation $(\mathcal{E}, U)$ is said to be completly reducible if there exists a basis $\mathcal{E}$ such that it is decomposable as a direct sum of irreducible representations.

6. So original :) !

When put in matrix notation, it means there exists a basis in which the action on $\mathcal{E}$ is

$$u(g) = \begin{pmatrix} u_1(g) & 0 & 0 & \dots & 0 \\ 0 & u_2(g) & 0 & \dots & 0 \\ 0 & 0 & u_3(g) & 0 & \vdots \\ \vdots & \vdots & 0 & \ddots & \vdots \\ 0 & 0 & \dots & \dots & \ddots \end{pmatrix}$$

and we naturally can't further reduce the blocks as we already are in the most "adequate" basis already.

The regular representation :
The regular representation $(\mathcal{E}, D)$ of a group $\mathcal{G}$, is defined as the representation constructed by making the matrices representing the $g_i \in \mathcal{G}$ such that, by constructing an orthonomal basis of the elements of $\mathcal{G}$ we have :

$$D(g_i)\,|g_j\rangle = |g_i g_j\rangle$$

and so

$$[D(g_k)]_{ij} = \langle g_i | D(g_k) | g_j \rangle$$

The regular representation's construction requires us to choose a basis, it is intermediary ad the states it wil physically act on aren't the elements of the symmetry groups naturally.

Additionnal Relevant information to physics

The irreducible representations of a group product is the tensor product of the group representations of the factors of said product.

The representation of a quotient group $\mathcal{G}/\mathcal{H}$ is the representation of $\mathcal{G}$ by composition with the natural homomorphism of $\mathcal{G}$ in $\mathcal{G}/\mathcal{H}$.

Let U be a representation of the group $\mathcal{G}$, U is not a representation of $\mathcal{G}/\mathcal{H}$ unless $\mathcal{H} \in \ker(U)$. If $\mathcal{H} \notin \ker(U)$ the representation U over $\mathcal{G}/\mathcal{H}$ is **multivalued**. A proof : for U to be a faithful representation of $\mathcal{G}/\mathcal{H}$ we would need to have

$$\forall g \in \mathcal{G}, \forall h \in \mathcal{H}, \qquad U(gh) = U(g)$$

and as U is a homomorphism we need $U(h) = e$ and so we need to have $\mathcal{H} \subseteq \ker(U)$ (linked to projective representations ?).

3. Schur's lemma

In Georgi's textbook [4] a neat proof for each part of this Lemma is substantiated, page 14.

α. Inequivalent representations

Schur's Lemma for inequivalent representations :

Let $(\mathcal{E}_1, U_1)$ and $(\mathcal{E}_2, U_2)$ be two inequivalent representations of $\mathcal{G}$, let T be a linear operator (intertwining operator) that maps from $\mathcal{E}_1$ to $\mathcal{E}_2$, if

$$TU_1(g) = U_2(g)T, \qquad \forall g \in \mathcal{G}$$

then $T = 0$, if the two reps were equivalent T could have been an isomorphism.

β. Equivalent representations

Schur's Lemma for equivalent representations :

Let $(\mathcal{E}, U)$ be a representation of $\mathcal{G}$, S be an invertible linear transformation, and T be an endomorphism on $\mathcal{E}$, if the following equation holds

$$TS^{-1}U(g)S = TU'(g) = U(g)T, \qquad \forall g \in \mathcal{G}$$

then $T \propto \lambda \mathbb{1}$ with $\lambda \in \mathbb{K}$.

$\mathbb{K}$ is the scalar field over which the vector space $\mathcal{E}$ is constructed.

4. Quantum Mechanics applications

In quantum mechanics as is reminded in [7] chapter 2, a symmetry transformation is a change of "point of view" that doesn't affect the outcome of possible measurements (basically conservation of inner products). According to Wigner's theorem any symmetry transformation has to be represented by a unitary and linear operator or by an anti-unitary and anti-linear operator.
A fundamental result that greatly impacts our use of GT and Reps in QM is the following :

Any unitary representation is completely reducible, and every symmetry transformations is thus completely reducible.

This result[7][8] follows from the study of the invariant subspace of the vector space $\mathcal{E}$ under the representation of the group $\mathcal{G}$ and its orthogonal complement.
Let us there be a vector space $\mathcal{E}$, a group $\mathcal{G}$, a unitary representation $(\mathcal{E}, D)$ of $\mathcal{G}$. Let us consider the representation to be reducible. From the previous consideration follows the existence of $\mathcal{V} \subset \mathcal{E}$ such that

$$\forall \left|v\right\rangle \in \mathcal{V}, \forall g \in \mathcal{G}, \qquad D(g)\left|v\right\rangle \in \mathcal{V}$$

Therefore if we consider $\mathcal{V}^\perp \subset \mathcal{E}$ such that

$$\forall \left|w\right\rangle \in \mathcal{V}^\perp, \qquad \langle w|v\rangle = 0$$

7. I didn't include a proof of the relation $\dim \mathcal{E} = \dim \mathcal{V} + \dim \mathcal{V}^\perp$, but it must hold.
8. I felt obligated to include this proof as I found it hard to find on the internet and in textbooks.

as $D(g^{-1})\,|v\rangle \in \mathcal{V}$, we get

$$\langle v|D(g)w\rangle = \langle D(g)^{\dagger}v|w\rangle = \langle v'|w\rangle = 0$$

therefore it follows that $D(g)\,|w\rangle \in \mathcal{V}^{\perp}$ and so $\mathcal{V}^{\perp}$ is also an invariant subspace of $(\mathcal{E}, D)$ of $\mathcal{G}$. We can write :

$$\mathcal{E} = \mathcal{V} \oplus \mathcal{V}^{\perp}$$

And so the complete reducibility of unitary representations, allows us to decompose our Hilbert space into N invariant subspaces of the group : $\mathcal{E} = \bigoplus_{i=1}^{N} \mathcal{E}_i$.
Let us consider the inequivalent invariant representation subspaces $\mathcal{E}_i$ and $\mathcal{E}_j$ and let us state that the simplest intertwining operator from $\mathcal{E}_i$ to $\mathcal{E}_j$ is the product of the projection operators over these two spaces P_i and P_j.
It follows naturally that $P_iU(t)P_j$ is also an intertwining operator and thusly

$$\left[P_iU(t)P_j\right] D_j(g) = D_i(g) \left[P_iU(t)P_j\right] \qquad \forall g \in \mathcal{G}$$

leads to $P_iU(t)P_j = 0$ according to Schur's Lemma. Physically this means that an eigenvector $|\psi(t)\rangle \in \mathcal{E}_k$ cannot evolve into a different irreducible representation over time but is bound to stay in the same vector sub-space $\mathcal{E}_k$.

Symmetries lead to selection rules.

5. Permutation group S_n

In physics it is very important to know the effect of permutations especially (if not exclusively) at the quantum level. Statistical Quantum Mechanics requires the postulates that emerge from the interaction of quantized fields, but that I do not yet know how to proof going from QFT, of symmetrization and anti-symmetrization.

The purely mathematical motivations are diverse too, but I do not find them specifically fascinating for the moment, so I shall not be writing them down for now.

The permutation group of n elements is written S_n. There are many notation conventions for the elements of S_n, I prefer the one in [4] as it is the simplest. The identity permutation is

$$(1)(2)(3)\cdots(n-1)(n)$$

another element could be

$$(16785)\cdots(n)$$

and the first cycle (16785) is a 5-cycle[9] that has the action of permutating the elements on which S_n acts in the following way :

$$x_1 \to x_6 \to x_7 \to x_8 \to x_5 \to x_1$$

9. Meaning there are 5 elements being acted on in a single cycle.

Any element of S_n has k_j j-cycles where :

$$\sum_{j=1}^{n} k_j j = n$$

The simplest representation of the permutation group, called **defining representation** according to Georgi's textbook, is built using a similar approach as the one used for the construction of the regular representation. We consider the elements being interchanged as the basis vectors. Then for a representation of $g_{k\leftrightarrow j} = (1)\cdots(kj)\cdots(n)$ we have :

$$U\,|k\rangle = |j\rangle$$

$$\langle l|U|k\rangle = \delta_{jl}$$

The conjugacy classes of S_n have only one thing to characterize them, the cyclic structure [10]. Note that an element of the permutation group is its own inverse, therefore a conjugacy class is built by multilying an element by another on both sides of it. More visually

$$g' = g g_1 g^{-1} = g g_1 g$$

Once again go to [4] pages 35-36 for "proofs" to the following statement, all conjugacy classes of S_n must have the same cycle structure.
Also the permutation (123) is the same as (231) and (12)(34) $\equiv$ (34)(12), I am implying by this that the number of element per conjugacy class N_{cc} is the number of elements of the same cyclic structure divided by the number of equivalent permutations per conjugacy class

$$N_{cc} = \frac{n!}{\Pi_j j^{k_j} k_j!}$$

6. Lie algebras

Algebra :

An algebra [11] is a vector space $(\mathcal{A}, \circ, \cdot)$ endowed with an additional binary operation $\star$ that is bilinear [12]

$$\star : \mathcal{A} \times \mathcal{A} \to \mathcal{A}$$
$$x, y \mapsto z \qquad x, y, z \in \mathcal{A}$$

bilinearity means of course

$$(x + y) \star z = x \star z + y \star z \qquad x \star (y + z) = x \star y + x \star z$$

10. Meaning the number of n-cycles and their order.
11. I'll use Fuchs' textbook [6] a lot for this section.
12. No other conditions needed, not even associativity or commutativity (this one we're used to not necessarily have).

Lie algebra :

A Lie algebra is an algeba[13] $(\mathfrak{g}, \circ, \cdot, [\,,])$ such that the bilinear operation, commonly denoted $[\,,]$, and called the **Lie bracket** satisfies, $\forall\, x, y, z \in \mathfrak{g}$:

$$[x, x] = 0 \quad \& \quad [x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$$

The second equation is called Jacobi's identity. The first equation simply reduces to $[x, y] = -[y, x]$ and is the **antisymmetry property**.
Making a Lie algebra from an associative algebra[14] is as simple as making up the commutator using the original multiplication

$$[x, y] := x\circ y - y\circ x, \qquad \forall x, y \in \mathcal{A}$$

Lie brakets are Lie algebra operations and are therefore conserved by homomorphisms. Let $\mathfrak{g}$ and $\mathfrak{h}$ be Lie algebras, let ϕ be a homomorphism from $\mathfrak{g}$ to $\mathfrak{h}$. Then

$$\begin{aligned} \phi : \quad & \mathfrak{g} \to \mathfrak{h} \\ & [x, y] \mapsto [\phi(x), \phi(y)] \qquad \forall x, y \in \mathfrak{g} \end{aligned}$$

α. Generalities

Generators :

Let $\mathfrak{g}$ be a Lie algebra, its dimension is the underlying vector space's dimension $\dim \mathfrak{g} = d$, if the vector space isn't uncountably infinite then we can write the basis of $\mathfrak{g}$

$$\mathcal{B} = \{T^a \mid a = 1, 2, \cdots, d\}$$

and the elements T^i are called the generators of the Lie algebra $\mathfrak{g}$. The name of these generators comes from the fact that the Lie braket opeation on generators of a Lie algebra characterize said algebraic structure uniquely with the **stucture constants** ${f^{ab}}_c$

$$[T^a, T^b] = {f^{ab}}_c T^c$$

A few lines on mappings from an algebraic structure to another :

- A **monomorphism** $\phi : \mathfrak{g} \to \mathfrak{h}$ is an injective mapping (for VS[15] $\ker(\phi) = \{0\}$).
- An **epimorphism** $\phi : \mathfrak{g} \to \mathfrak{h}$ is a surjective mapping (for VS $\mathrm{Im}(\phi) = \mathfrak{h}$).
- An **isomorphism** is both an epimorphism and a monomorphism.
- An **endomorphism** is a mapping ϕ such that $\phi : \mathfrak{g} \to \mathfrak{g}$.

13. Who could've guessed ?!
14. An algebra with an associative binary operation.
15. Vector spaces.

- An **automorphism** is an isomorphism and an endomorphism.
- A **derivation** is a mapping $\delta : \mathfrak{g} \to \mathfrak{g}$, such that

$$\delta([x, y]) = [\delta(x), y] + [x, \delta(y)]$$

$\forall x, y \in \mathfrak{g}$. One such derivation operator is the **adjoint map associated to** $x \in \mathfrak{g}$ defined by

$$\begin{aligned} \mathrm{ad}_x : \mathfrak{g} &\to \mathfrak{g} \\ y &\mapsto \mathrm{ad}_x(y) := [x, y] \end{aligned}$$

Lie subalgebras :

A Lie subalgebra [16] $(\mathfrak{h}, \circ, \cdot, [,])$ of a Lie algebra $(\mathfrak{g}, \circ, \cdot, [,])$ consists of a subspace $\mathfrak{h}$ of the vector space $\mathfrak{g}$ equipped with the same Lie braket as the one that endows $\mathfrak{g}$ with the Lie algebra status. Just as in [6], considering $\mathfrak{h}$ and $\mathfrak{l}$ two subsets of a Lie algebra $\mathfrak{g}$, we introduce the notation

$$[\mathfrak{h}, \mathfrak{l}] := \mathrm{span}_F \left\{ [T^a, T^b] \,|\, T^a \in \mathfrak{h},\, T^b \in \mathfrak{l} \right\}$$

where F stands for the *Field* over which the Lie algebra $\mathfrak{g}$ is defined. The meaning of all this is that any linear combination of the outputs of the Lie brackets of elements of $\mathfrak{h}$ and $\mathfrak{l}$, multiplied by elements of the Field, is an element of $[\mathfrak{h}, \mathfrak{l}]$.

According to the notation introduced just above, a Lie subalgebra of a Lie algebra $\mathfrak{g}$ is a subset $\mathfrak{n}$ of $\mathfrak{g}$ such that

$$[\mathfrak{n}, \mathfrak{n}] \subset \mathfrak{n}$$

Starting now [8] is also used as self-study material.

Ideals :

An ideal $\mathfrak{d}$ [17] of the Lie algebra $\mathfrak{g}$ is a subset of $\mathfrak{g}$, such that

$$[\mathfrak{g}, \mathfrak{d}] \subset \mathfrak{d}$$

$\mathfrak{d}$ is obviously a subalgebra of $\mathfrak{g}$. Put in words it is a set of elements of $\mathfrak{g}$ such that the span of their Lie brackets with any element of $\mathfrak{g}$ is still in that set of elements. It is a subset stable under Lie brackets with any element of the "mother-set".

Direct sums of Lie algebras :

If a Lie algebra $(\mathfrak{g}, \circ, \cdot, [,])$ is decomposable into a direct sum of n Lie algebras we write it

$$\mathfrak{g} = \bigoplus_{i=1}^{n} \mathfrak{g}_i$$

16. Some of the definitions are very logical and follow the usual logic for sub-*structure* but I never know what I could end up forgetting :p

17. In my mind it analogous to an invariant subgroup in some ways··· Do with this infotmation what you want future me !

If such Lie algebras follow a relation of the sort

$$[\mathfrak{g}_1, \mathfrak{g}_2] \subseteq \mathfrak{g}_1$$

on top of the following one

$$[\mathfrak{g}_i, \mathfrak{g}_j] = 0 \qquad i \neq j$$

We get $\mathfrak{g}$ a **semidirect sum**

$$\mathfrak{g} = \mathfrak{g}_1 \niplus \mathfrak{g}_2$$

If the direct sum is that of ideals $\mathfrak{g}_i$ of $\mathfrak{g}$ we denote it

$$\mathfrak{g} = \mathop{\boxplus}_{i=1}^{n} \mathfrak{g}_i$$

A Lie algebra which is decomposale into a direct sum of ideals and its components follows the relations ($[\ ,\ ]_i$ is the Lie braket of $\mathfrak{g}_i$)

$$[x, y] = [x, y]_i \qquad \forall\, x, y \in \mathfrak{g}_i \quad \forall i \in [\![1\,..\,n]\!]$$

$$[\mathfrak{g}_i, \mathfrak{g}_j] = 0 \qquad i \neq j$$

The derived algebra of a Lie algebra $\mathfrak{g}$ is

$$\mathfrak{g}' := [\mathfrak{g}, \mathfrak{g}] = \text{span}_F \{[x, y] \mid \forall x, y \in \mathfrak{g}\}$$

Its center is

$$\mathcal{Z}(\mathfrak{g}) = \{x \in \mathfrak{g} \mid [x, \mathfrak{g}] = 0\}$$

A **simple Lie algebra** is a non abelian Lie algebra with no proper ideals.
A **semisimple Lie algebra** is a Lie algebra that is the direct sum of simple Lie algebras.

Adjoint representation :

It is possible to reresent any Lie algebra $\mathfrak{g}$ on itself as it is a vector space, such a representation

$$\begin{aligned} U_{\text{ad}} : \ \mathfrak{g} &\to \text{GL}(\mathfrak{g}) \\ x &\mapsto \text{ad}_x \qquad \forall x \in \mathfrak{g} \end{aligned}$$

Thus the kernel consists of the center of the algebra, therefore it is an ideal. If the Lie algebra is simple, then the center is either the neutral element or $\mathfrak{g}$ itself, in the latter case the algebra is abelian because all elements of $\mathfrak{g}$ commute. Thusly the kernel is zero and the adjoint representation is a homomorphism (faithful).
If the Lie algebra is abelian, the adjoint representation is necessarily entirely made up of zeros, therefore it is not faithful.

β. Cartan Subalgebra

A Cartan subalgebra is "built".
First, one needs to consider the generators of a Lie algebras $x \in \mathfrak{g}$ [18] to make the ad_x map diagonalizable. We take the maximal set of commuting [19] hermitian operators that are linearly independent, to diagonalize as much as we can. This set being chosen in the Lie algebra $\mathfrak{g}$, it is a subset of $\mathfrak{g}$.

If we take the generators of the abelian subalgebra (called semisimple generators/elements meaning diagonalizable) chosen such that

$$[H^i, H^j] = 0 \qquad \forall\, i, j \in [\![1\,..\,r]\!]$$
$$H^i = (H^i)^\dagger$$

The resulting

$$\mathfrak{g}_0 := \mathrm{span}_{\mathbb{C}} \left\{ H^i \,|\, \forall\, i, j = 1, 2, \cdots, r \right\}$$

is called the **Cartan subalgebra**.
The field over which it is constructed over is $\mathbb{C}$ because it is an algebraically closed field, any equation of $\mathbb{C}$ has solutions in $\mathbb{C}$ [20], more on that in Fuchs' textbook page 83.
The integer r is the rank of $\mathfrak{g}$

$$r = \mathrm{rank}\ \mathfrak{g} = \dim \mathfrak{g}_0.$$

γ. Roots and Cartan-Weyl basis

Remember that algebras are vector spaces. Implying that the generators are vectors of this VS.
Using the adjoint representation it is possible to construct a basis on which the representations of the Cartan generators are "maximally diagonal", such basis vectors (generators of the Lie algebra on which the Cartan generators act) are called **step operators/generators**.

To put it differently, since the H^is all commute, so do the adjoint representations adj_{H^i}, moreover we can find a basis of *Step operators* which are also generators of $\mathfrak{g}$, such that the set of H^is is diagonal, they are indeed an abelian subset of hermitian matrices, they are simultaneously diagonalizable.

Applying the adjoint representation of the cartan generators h on the elements of the Lie algebra y leads to

$$\mathrm{adj}_h(y) = [h, y] = \alpha_y(h) y$$

where $\alpha_y(h)$ is an element of the dual vector space $\mathfrak{g}_0^\star$ to the vector space $\mathfrak{g}_0$. These $\alpha_y(h)$ depend linearly on their respective h.
As usual in Linear algebra (here we focus on the vector space structure of the Lie algebra so linear algebra applies), the eigenvalues of adj_h are the *roots* of the characteristic polynomial equation of

18. Also written $T_a \in \mathfrak{g}$
19. Commuting set of hermitian operators means an abelian subalgebra of hermitian elements of said algebra
20. $x^2 = -1$ doesn't have a solution in $\mathbb{R}$ for example.

h.
The function α_h is therefore called a **root** of the Lie algebra $\mathfrak{g}$[21].

The Lie algebra $\mathfrak{g}$ is spanned by the step operators AND the cartan generators, so we can write

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha \neq 0} \mathfrak{g}_\alpha \tag{6.1}$$

where $\mathfrak{g}_\alpha = \left\{ x \in \mathfrak{g} \mid [h, x] = \alpha(h)x,\ \forall h \in \mathfrak{g}_0 \right\}$. Or

$$\mathfrak{g} = \bigoplus_{\alpha} \mathfrak{g}_\alpha$$

the sum is simply a direct sum of subalgebras, not a direct sum of ideal algebras denoted $\boxplus$.

The Equation 6.1 is called the **root space decomposition of** $\mathfrak{g}$ relative to the Cartan subalgebra $\mathfrak{g}_0$, and we usually write the previous equations

$$[H^i, E^\alpha] = \alpha^i E^\alpha, \qquad \forall i \in [\![1\,..\,r]\!]$$

and where α^i is the i-th component of the root vector previously called root of $\mathfrak{g}$. Which are vectors of the dual vector space to the Cartan subalgebra $\mathfrak{g}_0$. The set

$$\mathcal{B} = \left\{ H^i \mid i \in [\![1\,..\,r]\!] \right\} \cup \{ E^\alpha \mid \alpha \in \Phi \} . \tag{6.2}$$

is the **Cartan-Weyl basis**. $\Phi = \Phi(\mathfrak{g})$ is the set of all roots of $\mathfrak{g}$ called **root system** of $\mathfrak{g}$.

δ. Killing form

The inner product used to further study the roots of $\mathfrak{g}$, defined on the root space $\mathfrak{g}_O^\star$ and by analogy on its dual space $\mathfrak{g}_0$, which is itself the restriction of an inner product on $\mathfrak{g}$. This product is a bilinear map from $\mathfrak{g} \times \times \to \mathbb{C}$, and is called the **Killing Form** κ of $\mathfrak{g}$.
For finite-dimensional Lie algebras it is defined by the trace on the adjoint representation

$$\kappa(x, y) := \mathrm{tr}\,(\mathrm{adj}_x \circ \mathrm{adj}_y), \qquad \forall\ x, y \in \mathfrak{g} \tag{6.3}$$

As we already know it, the trace is a linear and symmetric map $\kappa(x, y) = \kappa(y, x)$.

By defining the adjoint of a Lie bracket as

$$\mathrm{adj}_{[x,y]} = \mathrm{adj}_x \circ \mathrm{adj}_y - \mathrm{adj}_y \circ \mathrm{adj}_x \tag{6.4}$$

A property of « invariance » of the Killing form follows

$$\kappa([x, y], z) = \kappa(x, [y, z]) \tag{6.5}$$

21. We didn't properly exclude the Cartan generators from the Step operators so a lot of the roots are 0, exactly $r =_t extrank\mathfrak{g}_0$ of the roots will be ignored, we wil therefore ignore those and only call roots the non-zero roots. Also, the roots are relative to the Cartan basis chosen.

In a basis $\mathcal{B} = \{T^a \mid a = 1, \cdots, d\}$ the Killing forms represented by a matrix whose components are

$$\kappa^{ab} := \frac{1}{C_{\text{ad}}}\kappa(T^a, T^b) = \frac{1}{C_{\text{ad}}}\operatorname{tr}\left(\text{adj}_{T^a}\circ\text{adj}_{T^b}\right)$$

C_{ad} is a normalization constant.
Cartan proved that the Killing form is non-degenerate when the Lie algebra is finite-dimensional and semi-simple, and is therefore a proper inner product as opposed to a trivial one where there may be a $x \in \mathfrak{g},\ x \neq 0$ such that $\kappa(x, y) = 0,\ \forall y \in \mathfrak{g}$. It is also non-degenerate when restricted to the Cartan subalgebra $\mathfrak{g}_0$.

Any non-degenerate bilinear form on a VS can be used to identity the VS and its dual. Put differently any H^α of $\mathfrak{g}_0$ can be associated to a root α of $\mathfrak{g}_0^\star$, such that

$$\alpha(h) = c_\alpha \kappa(H^\alpha, h), \qquad \forall h \in \mathfrak{g}_0$$

Using the Killing form one can finally define a non-degenerate inner product on $\mathfrak{g}_0^\star$ by setting

$$(\alpha, \beta) := c_\alpha c_\beta \kappa(H^\alpha, H^\beta) = c_\beta \alpha(H^\beta)$$

for all roots α, β, by bilinearity it can be extended to the whole $\mathfrak{g}_0^\star \times \mathfrak{g}_0^\star$.

IV Differential geometry

I will use Nakahara's textbook [9] for the first two sections mainly, accompanied by Franck Pacard's lecture notes for the third section and forward, lecture notes on which my upcoming class is based.

1. Basics

Here are some relevant defintions reformulated to be as clear as possible.

Maps :

A map, or mapping, is a rule by which, for two sets X and Y, we assigne to $x \in X$ an element $y \in Y$

$$\begin{aligned} f: \quad X &\to Y \\ x &\mapsto y \end{aligned}$$

Most people with a high school degree already know it but here it comes, it is often denoted

$$f: \quad x \mapsto f(x) = y$$

The one-to-one correspondance of these elements x and y is however not guaranteed, so the concept of **inverse image** is proposed to be

$$f^{-1}(y) = \{x \in X \mid f(x) = y\}$$

And so, three kinds of mappings exist :

- → **Injective mappings** are the ones such that the inverse image of an element $y \in Y$, either exists and is unique or doesn't exist.
- → **Surjective mappings** are the ones such that for any

$$x \in X,\ \exists \text{ at least one } y \in Y, \text{ s.t } f(x) = y.$$

- → **Bijective mappings** are Injective and Surjective, and have one-to-one mapping, so bijective mappings allow for « structure preserving rules ».

Equivalence relations :

An equivalence relation $\sim$ is a relation between two elements of a set $(a, b) \in X^2$, that has the following propreties

$$a \sim a, \qquad a \sim b \implies b \sim a, \qquad a \sim b \text{ and } b \sim c \implies a \sim c$$

Equivalence classes :

Such mathematical objects are defined by an element of the set we work on

$$[a] = \{x \in X \,|\, x \sim a\}$$

A simple proof based on the propreties of $\sim$ allows us to see how equivalence relation $\sim$ divide the set X on which it is defined as mutually disjoint subsets of X. It is a consequence of the transitivity, symmetry and commutation of equivalence relations.

The notion of dual vector space is essential to understanding tensors, so here is a re-introduction to the basic concepts to know [1].

Dual vector spaces :

A dual vector space is defined as the **dual** of a vector space ; consisting of mappings from the vector space it is dual to, to the field on which the « original » vector space is constructed. So for a vector space $\mathcal{V}$, the dual vector space $\mathcal{V}^*$ is

$$\mathcal{V}^* = \mathrm{span}_{\mathbb{K}}\,\{f(\cdot)\,|\,\text{s.t } \forall\, v \in \mathcal{V},\ f(v) \in \mathbb{K}\}$$

The elements e_i are the basis elements of $\mathcal{V}$, and the action of f on v is the following

$$f(v) = v^i f(e_i) \equiv \langle f, v\rangle$$

where

$$\begin{aligned} \langle\,.\,,\,.\,\rangle :\ & \mathcal{V}^* \times \mathcal{V} \to \mathbb{K} \\ & f\ ,\ v \mapsto k \in \mathbb{K} \end{aligned}$$

Those fs are linear and can be decomposed as $\boldsymbol{f} = f^*_i e^{*i}$, where the e^{*i} are the basis elements of the dual vector space. The elements of the dual VS are therefore also vectors, called **dual vectors**, they indeed verify the axioms we defined VSs with. We always choose dual vectors so that

$$e^{*i} e_j = \delta^i{}_j.$$

The vector components in a basis f^*_i and f^i are called **covariant** and **contravariant** components respectively, any index that is up is a contravariant index and any lower index is a covariant index.

Tensors

1. Gilles Abramovici's lecture notes are exceptionally clear and were very pleasant to read

A tensor of ank (p, q) is a multilinear map that maps p dual vectors and q vectors to the scalar field of the vector spaces associated

$$T : \bigotimes^{p} \mathcal{V}^* \bigotimes^{q} \mathcal{V} \to \mathbb{K}$$

2. Topology

Topology is the field of study of global structures and measures.
All of the things you define in Topology are defined in tems of open sets, so they all depend on how you define open sets.

Topological spaces and topologies :

Let X be a set, and let

$$\mathcal{T} = \left\{U_i \,|\, U_i \in X,\ i \in I\right\},$$

where $\mathcal{T}$ is a collection of subsets of X. $(X, \mathcal{T})$ is said to be a topological space if :

i) $\emptyset, X \in \mathcal{T}$.
ii) If J is *any*[2] subcollection of I, then the set of subsets $\left\{U_j\right\}_{j\in J}$ satisfies $\cup_{j\in J} U_j \in \mathcal{T}$.
iii) If K is *any finite*[3] subcollection of I, then the set of subsets $\left\{U_k\right\}_{k\in K}$ satisfies $\bigcap_{k\in K} U_k \in \mathcal{T}$.

In case $(X, \mathcal{T})$ is a topological space, $\mathcal{T}$ is what endows X with its topology, and is simply called the **topology** of X, and the U_is are called **open sets**.

Metric

A metric d is a mapping

$$d :\ X \times X \to \mathbb{K}$$

satisfying the following conditions
(i) $d(x, y) = d(y, x)$.
(ii) $d(x, y) \geq 0$, $d(x, y) = 0 \implies x = y$.
(iii) $d(x, y) + d(y, z) \geq d(x, z)$.
where $\forall\, x, y, z \in X$.

Continuous maps

A continuous map is a map from one topological space X to a topological space Y, such that the inverse of an open set in Y is an open set in X[4].

2. may be infinite.
3. may not be infinite.
4. Check out *Geometry, Topology and Physics* by M.Nakahara first edition page 50 for explanations as to why this choice holds up.

Neighbourhood

Let $(X, \mathcal{T})$ be a topological space, the neighbourhood N of an element $x \in X$, is a subset of X containing an open subset U_i to which x belongs.

Hausdorff space

A Hausdorff space is a topological space $(X, \mathcal{T})$ such that for any two elements $x, x' \in X$, there always exists repsective neighbourhoods such that $U_x \cap U_{x'} = \emptyset$.

A **closed subset** A of $(X, \mathcal{T})$ is a subset such that its complement $X - A \in \mathcal{T}$ is open (meaning in $\mathcal{T}$).

Let A be a subset of a topological space here are three of its interesting propreties :

→ Its **closure**, written $\overline{A}$, is the smallest closed set that contains A.

→ Its **interior**, written A° is the largest open subset of A.

→ Its **boundary**, $b(A) = \overline{A} - A^\circ$, is the complement of A° in $\overline{A}$.

A **covering** of a topological space $(X, \mathcal{T})$ is a set of subsets $\{A_i\}$ that realize the condition

$$\bigcup_{i \in I} A_i = X$$

We would call it an **open covering** if all the A_i were open.

Compactness

Consider a topological space $(X, \mathcal{T})$, it is said that the topological space is compact if for every open covering $\{U_i \mid i \in I\}$ of X, there exists a *finite* subcollection J of I such that the sets $\{U_j \mid j \in J\}$ are also an open covering of X.

Connectedness

A topological space is said to be **connected** if it *cannot* be written as $X = U_1 \cup U_2$ and these subsets satisfy $U_1 \cap U_2 = \emptyset$.
A topological space $(X, \mathcal{T})$ is said to be **arcwise connected** if $\forall\, x, y \in X,\ \exists f : [0, 1] \to X$, with f a continuous map and $f(0) = x$ & $f(1) = y$.
A loop in $(X, \mathcal{T})$ is a continuous map $f : [0, 1] \to X$ with $f(0) = f(1)$. If any such loop can be continuously shrunk into a point, X is called **simply connected**.

Homeomorphism

A homeomorphism is a continuous invertible map from a topological space X_1 to another topological space X_2 such that its inverse is also continuous

$$\begin{aligned} \phi : X_1 &\to X_2 \\ x &\mapsto y \end{aligned}$$

3. Manifolds

Now as stated earlier in the document, this section is based on two sources Frank Pacard's lecture notes on differential geometry and Marios Petropoulos's lectue notes on GR.

Topological manifolds

An m-dimensional topological manifold is a topological space $(X, \mathcal{T})$ that has $\forall x$ an open subset $U_x \in \mathcal{T}$ containing x, that is homeomorphic to V_x an open subset of $\mathbb{R}^m$, the topological space is also a Hausdorff space and is *second countable*[5].

The previous definition is very rigorous, in Nakahara's [9], the definition is that of a smooth manifold directly, I guess it is because it is the definition most relevant to physics (?).

Chart

Let us consider an m-dimensional manifold M, a local chart is a triplet (ϕ, U, V), where U is an open subset of M, V is an open subset of $\mathbb{R}^m$, and $\phi : U \to V$ is the homeomorphism linking them.

Atlas

An atlas is a collection of charts $(\phi_\alpha, U_\alpha, V_\alpha)$, $\alpha \in A$ such that $\bigcup_{\alpha \in A} U_\alpha$ covers M.

4. Differential forms

5. Riemannian Geomerty

6. Fiber Bundles

5. Meaning it has a countable open set basis.

Exercises

V MAT553

1. Chapter 1

α. Exercise 1.1

$\rightarrow$ *Prove that a compact metric space (X, d) is second countable.*

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